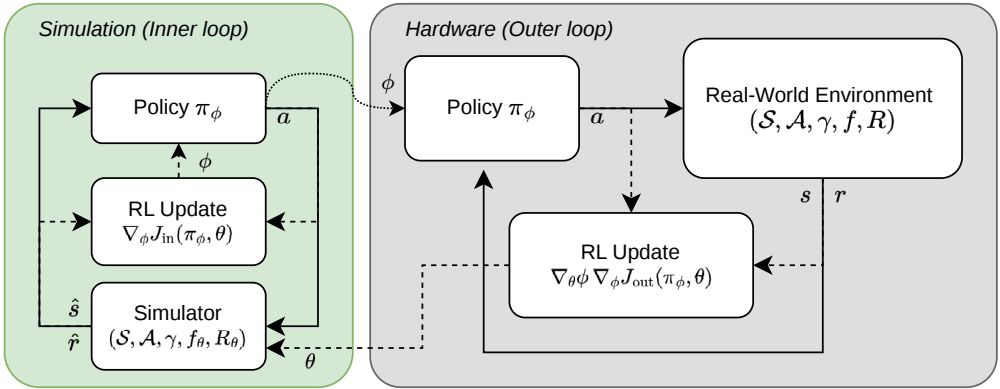
Simulation (Inner loop)
Policy π_ϕ
a
ϕ
RL Update
$\nabla_\phi J_{\text{in}}(\pi_\phi, \theta)$
$\hat{s}$
$\hat{r}$
Simulator
$(\mathcal{S}, \mathcal{A}, \gamma, f_\theta, R_\theta)$
θ
Hardware (Outer loop)
ϕ
Policy π_ϕ
a
Real-World Environment
$(\mathcal{S}, \mathcal{A}, \gamma, f, R)$
s
r
RL Update
$\nabla_\theta \phi \nabla_\phi J_{\text{out}}(\pi_\phi, \theta)$